Day-ahead layer
(once per day, Section 4.1)

input

Historical days Ω^{DA}

load

hour

14 weekdays: computing, inference, weather

lines 1–2

Closed-loop back-test

day d

commitment $C \rightarrow$

intraday controller on every day, $C \in \mathcal{G}$ until first failure

line 3

Capability per day

$\hat{F}_N$

ε_N

capability ξ

capability ξ_d (47); $\hat{P}_N$, radius ε_N (53)

line 4

Sorted-margin certificate

$\delta_{(j)}(C)$

$\Phi_\rho(C)$

days sorted by margin

knapsack (50), dual (51): $\Phi_\rho(C) \geq \varepsilon$ (52) (Prop. 4)

line 5

Cost-optimal commitment

$\tilde{J}(C)$

C^{DRO}

C^*

commitment C

C^* (55): least back-tested cost below $C_\rho^{DRO}(\varepsilon)$ (48)

the same controller is back-tested

commitments C_k, terminal references

Intraday layer
(every 15 min, Section 4.2)

line 8

Measure state

B_1 B_2 B_3 E S

$\hat{\mathbf{x}}_t$: backlogs, stored energy and cold, gas engine, events

line 9

Forecasts and scenarios

now

time

error-corrected forecasts (JR-D) or a scenario tree (JR-S)

line 10

Cross-event joint recovery MPC

horizon $[t,\ t+T_H]$

catch-up / recharge

$t+\Delta t$

t E1 E2

14 18 22 26 30 h

MILP (58) over the 18-h horizon with all confirmed events: which debt to repay first

lines 11–12

Apply and settle

$0.8C_k$ C_k $1.2C_k$

$\bar{r}_k$

valid full cap

first step applied, plant replay, EDF execution; settlement (19–22)

$t \leftarrow t + \Delta t$